

Laplace Domain

Control Engineering

Course Schedule

1. Systems
2. Continuous-time Controllers
3. Laplace Domain
4. Poles/Zeros & Stability
5. Controller Design
6. Discrete-time Controllers
7. State Space Representation
8. Advanced Control Methods

Laplace Domain

- Laplace transform
- Transfer function
- Frequency response
- Bode plots
- Bandwidth

Why the Laplace Transform?

- Converts differential equations (dynamic systems) into algebraic equations → easier to solve
- Simplifies analysis and design
- Natural language of control theory

Reminder: Fourier Transform

Transformation from time to frequency domain:

$$\mathcal{F}\{f(t)\} = F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-i\omega t} dt$$

Tells which frequencies ω are present in a signal or system

To be exact, tabulation of both magnitude and phase response at each frequency $\rightarrow$ complex numbers:

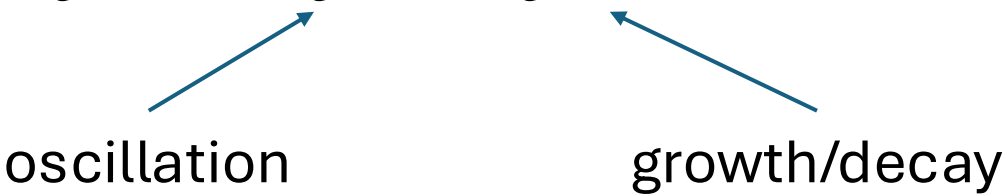
$$F(\omega) = |F(\omega)|e^{i\phi}$$

- $|F(\omega)|$: how much of that frequency in the signal (how much the system scales each frequency)
- ϕ : time shift of that frequency (how much the system shifts each frequency)

Idea of Laplace Transform

Extend Fourier transform to include information about growth/decay behavior

→ Add exponential weighting:

$$e^{-i\omega t} \rightarrow e^{-i\omega t} \cdot e^{-\sigma t}$$


oscillation growth/decay

Combine frequency ω and growth/decay rate σ :

$$\omega \rightarrow s = \sigma + j\omega$$

Definition of Laplace Transform

Integral transform from time to complex frequency s :

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^{\infty} f(t)e^{-st} dt$$

$$s = \sigma + j\omega$$

Back transformation:

$$\mathcal{L}^{-1}\{F(s)\} = f(t) = \frac{1}{2\pi j} \int_{\sigma - \infty}^{\sigma + \infty} F(s)e^{st} ds$$

Example

Laplace transform of t :

$$\mathcal{L}\{t\} = \int_0^{\infty} t \cdot e^{-st} dt$$

Using partial integration: $\int u dv = uv - \int v du$

$$u = t \Rightarrow du = dt$$

$$dv = e^{-st} dt \Rightarrow v = -\frac{1}{s} e^{-st}$$

$$\Rightarrow \int_0^{\infty} t \cdot e^{-st} dt = \underbrace{\left[-\frac{t}{s} e^{-st} \right]_0^{\infty}}_{=0} + \int_0^{\infty} \frac{1}{s} e^{-st} dt = -\frac{1}{s^2} [e^{-st}]_0^{\infty} = \frac{1}{s^2}$$

$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases}$$

Laplace of Common Signals

Function	Time domain	Laplace domain
Unit Step	$u(t)$	$\frac{1}{s}$
Delayed unit step	$u(t - \tau)$	$\frac{1}{s} e^{-\tau s}$
Unit impulse	$\delta(t)$	1
Delayed unit impulse	$\delta(t - \tau)$	$e^{-\tau s}$
Ramp	$t \cdot u(t)$	$\frac{1}{s^2}$
Exponential decay	$e^{-\alpha t} \cdot u(t)$	$\frac{1}{s + \alpha}$
Sine	$\sin \omega t \cdot u(t)$	$\frac{\omega}{s^2 + \omega^2}$
Cosine	$\cos \omega t \cdot u(t)$	$\frac{s}{s^2 + \omega^2}$

Properties of Laplace Transform

Linearity: $\mathcal{L}\{a \cdot f_1(t) + b \cdot f_2(t)\} = a \cdot F_1(s) + b \cdot F_2(s)$

Convolution: $\mathcal{L}\{f_1 * f_2\} = \mathcal{L}\left\{\int_0^t f_1(\tau) \cdot f_2(t - \tau) d\tau\right\} = F_1(s) \cdot F_2(s)$

Derivative: $\mathcal{L}\left\{\frac{df(t)}{dx}\right\} = sF(s)$

Integration: $\mathcal{L}\left\{\int_0^t f(\tau) d\tau\right\} = \mathcal{L}\{(u * f)(t)\} = \frac{1}{s} F(s)$

Frequency Shifting: $\mathcal{L}\{e^{at} \cdot f(t)\} = F(s - a)$

Time Shifting: $\mathcal{L}\{f(t - a) \cdot u(t - a)\} = e^{-as} \cdot F(s)$

Time Scaling: $\mathcal{L}\{f(at)\} = \frac{1}{a} F\left(\frac{s}{a}\right)$

From Differential Equation to Transfer Function

Example: PT2 system

Time domain:

$$u(t) \longrightarrow \boxed{\ddot{y}(t) + 2\zeta\omega_n\dot{y}(t) + \omega_n^2y(t) = \omega_n^2Ku(t)} \longrightarrow y(t)$$

Frequency domain:

$$U(s) \longrightarrow \boxed{s^2Y(s) + 2\zeta\omega_nsY(s) + \omega_n^2Y(s) = \omega_n^2KU(s)} \longrightarrow Y(s)$$

$$\text{Transfer function: } G(s) = \frac{Y(s)}{U(s)} = \frac{\omega_n^2 K}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (\text{assuming zero initial conditions})$$

Transfer Function

Ratio of output $Y(s)$ to input $U(s)$ in Laplace domain

Fully describes input–output dynamics of system, independent of input

$$Y(s) = G(s) \cdot U(s)$$

In time domain, this corresponds to the convolution of the input with the impulse response of the system, **assuming zero initial conditions**:

$$y(t) = \int_0^t g(t - \tau) \cdot u(\tau) d\tau$$

Transfer Function for PT1 System

Differential equation:

$$\tau \cdot \dot{y}(t) + y(t) = K \cdot u(t)$$

Laplace transform:

$$\tau \cdot s \cdot Y(s) + Y(s) = K \cdot U(s)$$

Transfer Function:

$$G(s) = \frac{Y(s)}{U(s)} = \frac{K}{1+\tau s} \quad \leftarrow \text{independent of input}$$

Example: Free Fall with Viscous Damping

$$m \cdot \dot{v}(t) + c \cdot v(t) = m \cdot g$$

$$\text{where } u(t) = \begin{cases} 0, & t < 0 \\ m \cdot g, & t \geq 0 \end{cases} \rightarrow U(s) = m \cdot g \cdot \frac{1}{s}$$

$$m \cdot s \cdot V(s) + c \cdot V(s) = m \cdot g \cdot \frac{1}{s}$$

$$V(s) = \frac{m \cdot g}{s(m \cdot s + c)} = \frac{g}{s \left(s + \frac{c}{m} \right)}$$

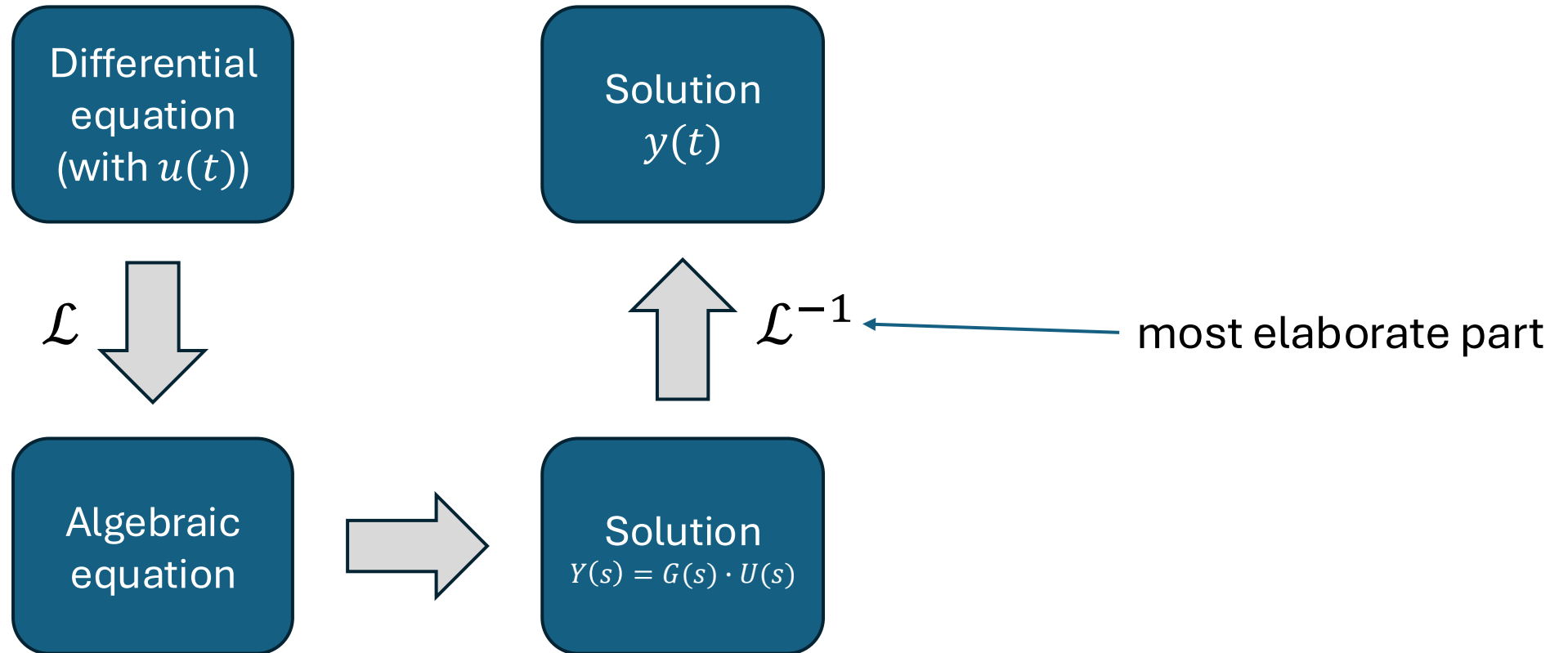
Transformation table:

$$\frac{\alpha}{s(s + \alpha)} \mapsto 1 - e^{-\alpha t}$$

$$\rightarrow v(t) = g \cdot \frac{m}{c} \cdot \left(1 - e^{-\frac{c}{m}t} \right)$$

$$\tau = K = \frac{m}{c}$$

System Response with Transfer Function



Transfer Function in Polynomial Form

$$G(s) = \frac{K}{1 + \tau s}, \quad G(s) = \frac{K}{\frac{1}{\omega_n^2} s^2 + \frac{2\zeta}{\omega_n} s + 1}, \quad \dots$$

Input coupling:
describes how the input enters
the system (can contain time
derivatives)

$$\rightarrow G(s) = \frac{b_m s^m + \dots + b_1 s + b_0}{a_n s^n + \dots + a_1 s + a_0} = \frac{b_0}{a_0} \frac{\frac{b_m}{b_0} s^m + \dots + \frac{b_1}{b_0} s + 1}{\frac{a_n}{a_0} s^n + \dots + \frac{a_1}{a_0} s + 1}$$

fundamental theorem
of algebra



$$K = \frac{b_0}{a_0}$$

Intrinsic system dynamics:
characteristic polynomial of the
system

$$\rightarrow G(s) = \frac{b_m}{a_n} \frac{\prod_{i=1}^m (s - s_{0i})}{\prod_{i=1}^n (s - s_{pi})}$$

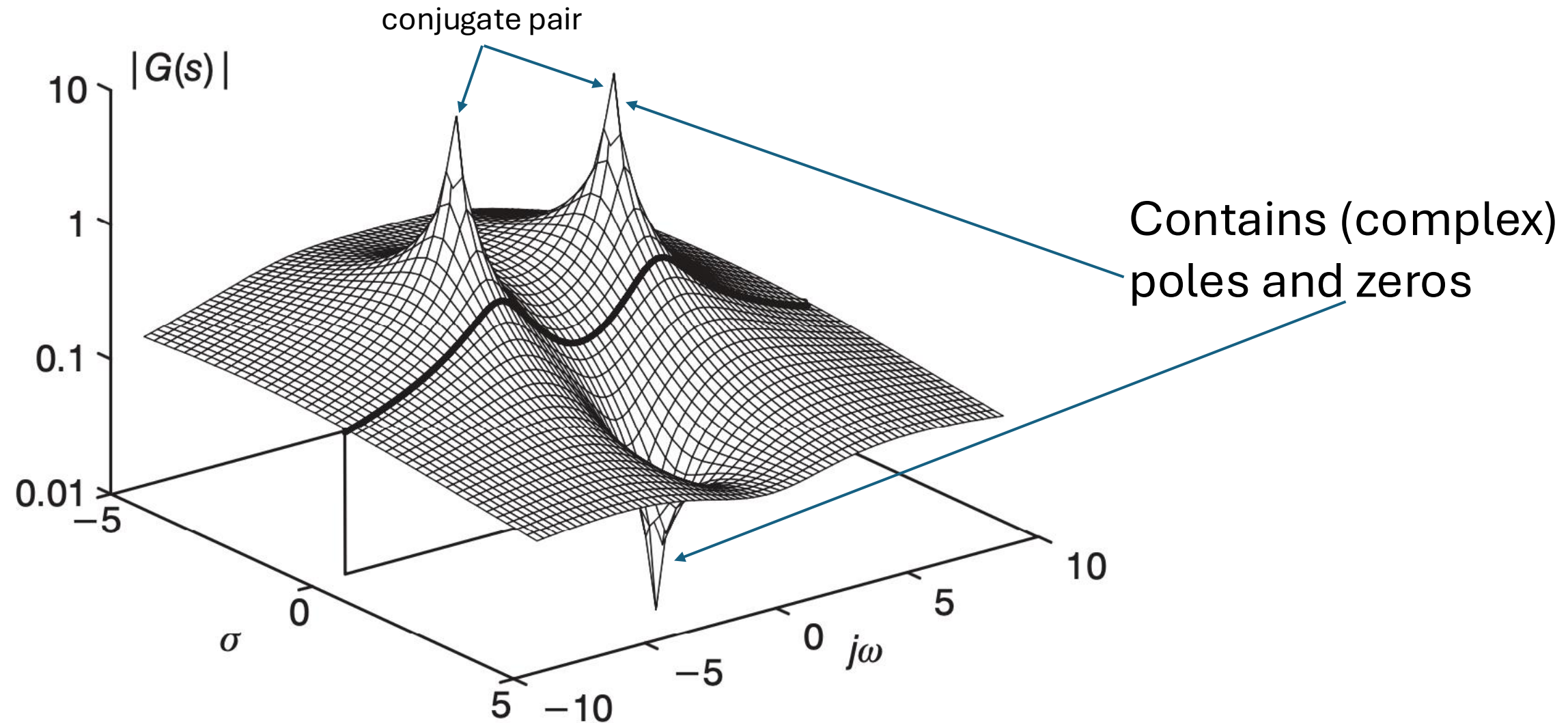
roots of numerator: m zeros s_0

roots of denominator: n poles s_p

$$k = \frac{b_m}{a_n}$$

Causality requires $n \geq m$ (output cannot depend on derivative of input)

Graphical Interpretation of Transfer Function



Poles and Zeros

Poles and zeros of a transfer function can be real or complex

Complex poles/zeros always appear in conjugate pairs for real physical systems (need for real coefficients in differential equations)

$$(s - (\sigma + j\omega))(s - (\sigma - j\omega)) = (s - \sigma)^2 + \omega^2$$

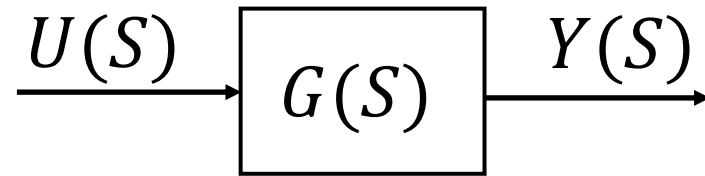
Poles determine stability and dynamics (system's natural response)

$$y_i(t) = C_i e^{s_{pi}t}, \quad y_{\text{hom}}(t) = \sum_i y_i(t)$$

More about this later ...

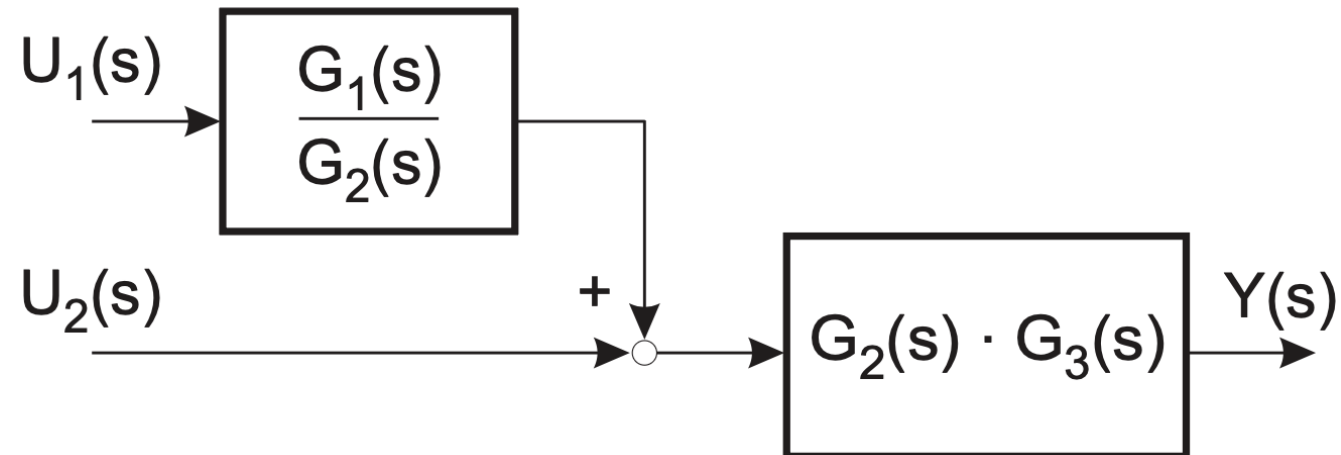
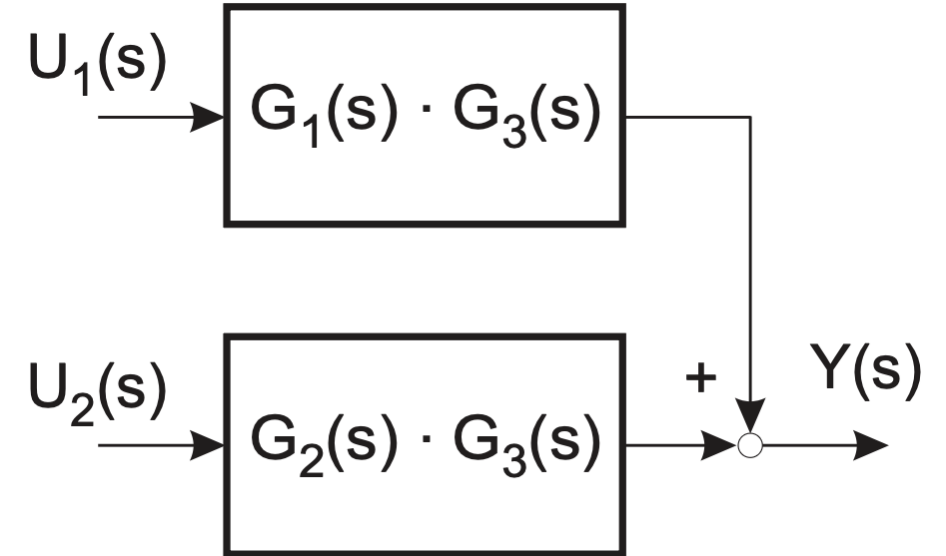
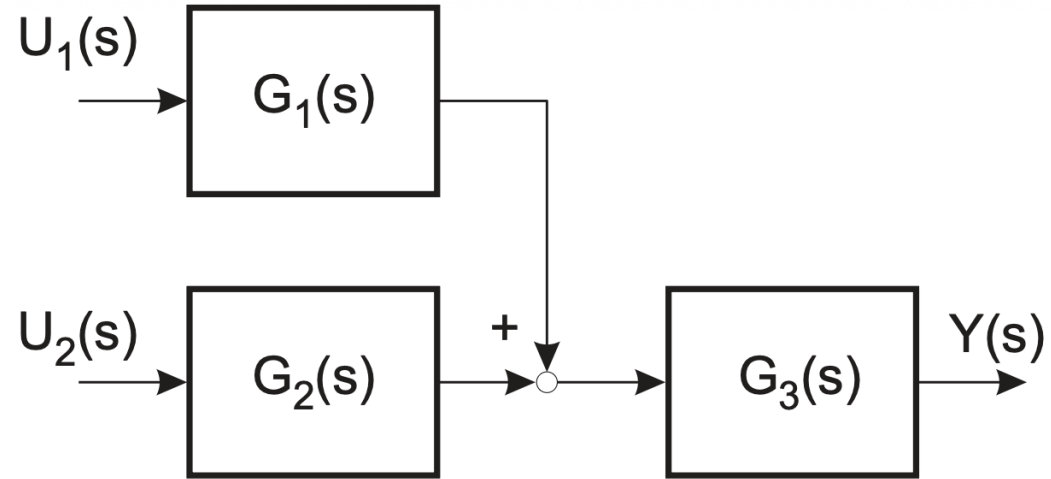
Zeros shape the forced (input-driven) response

Block Diagrams

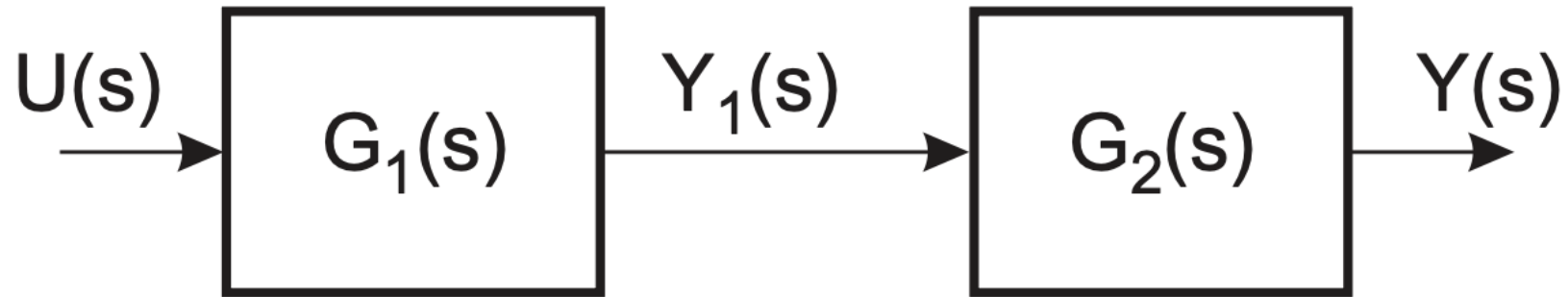


Modular representation

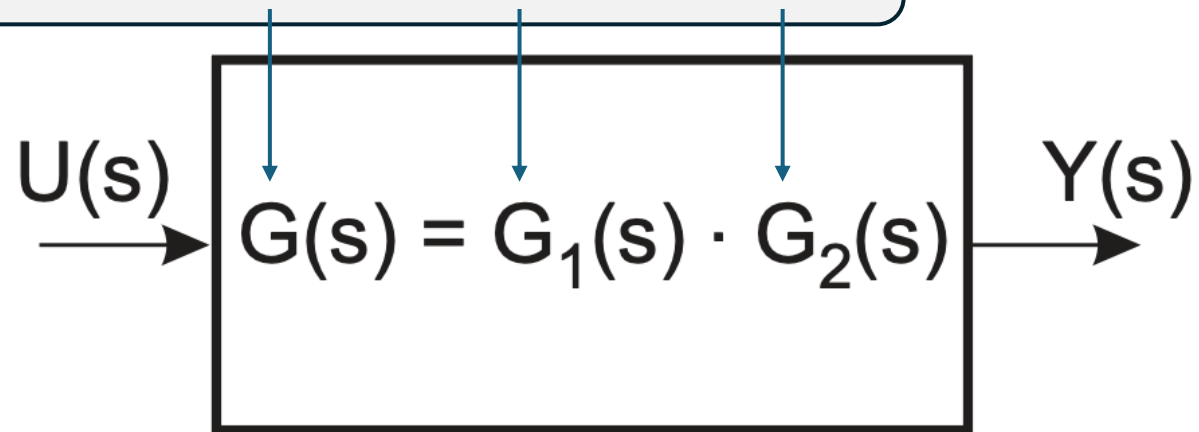
Block Diagram Algebra



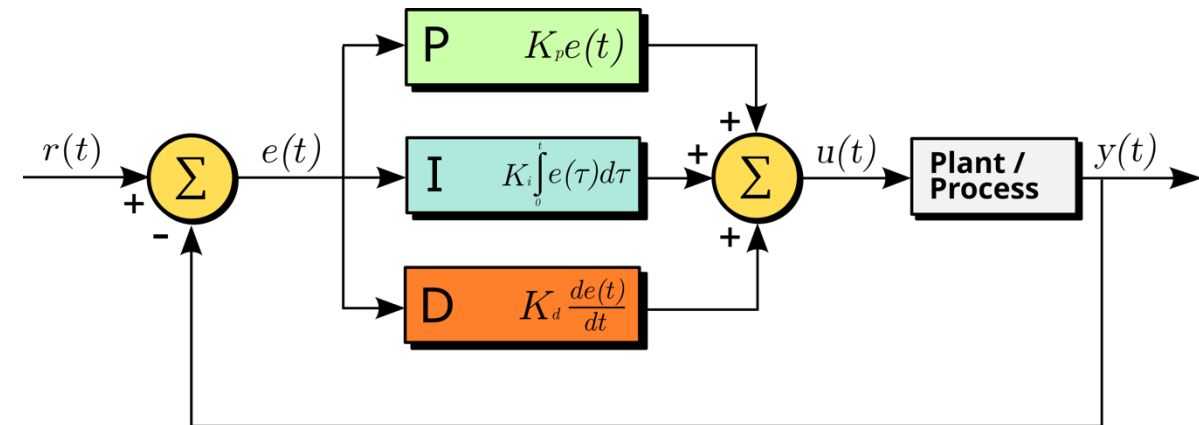
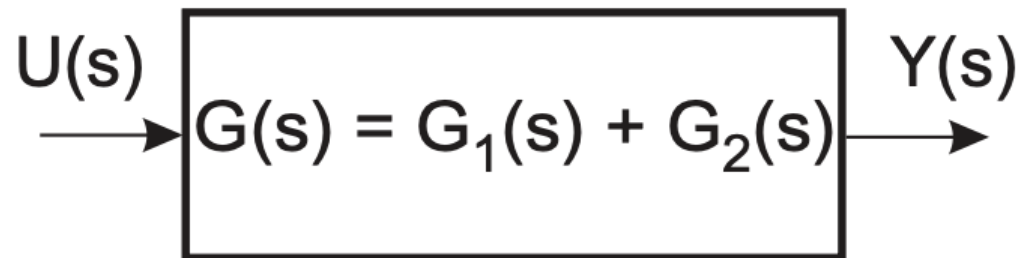
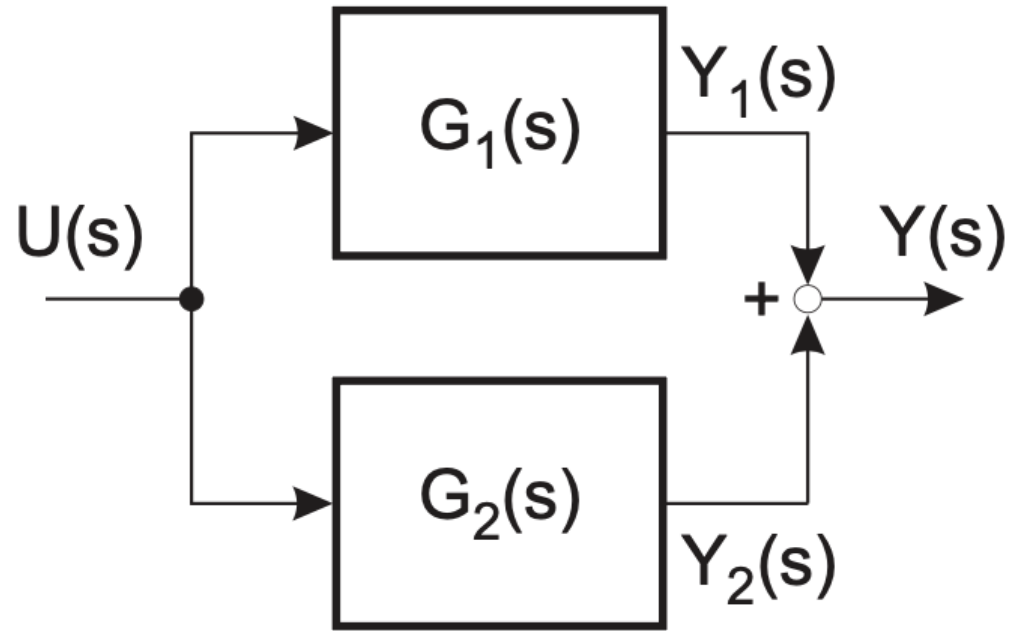
Series Connection



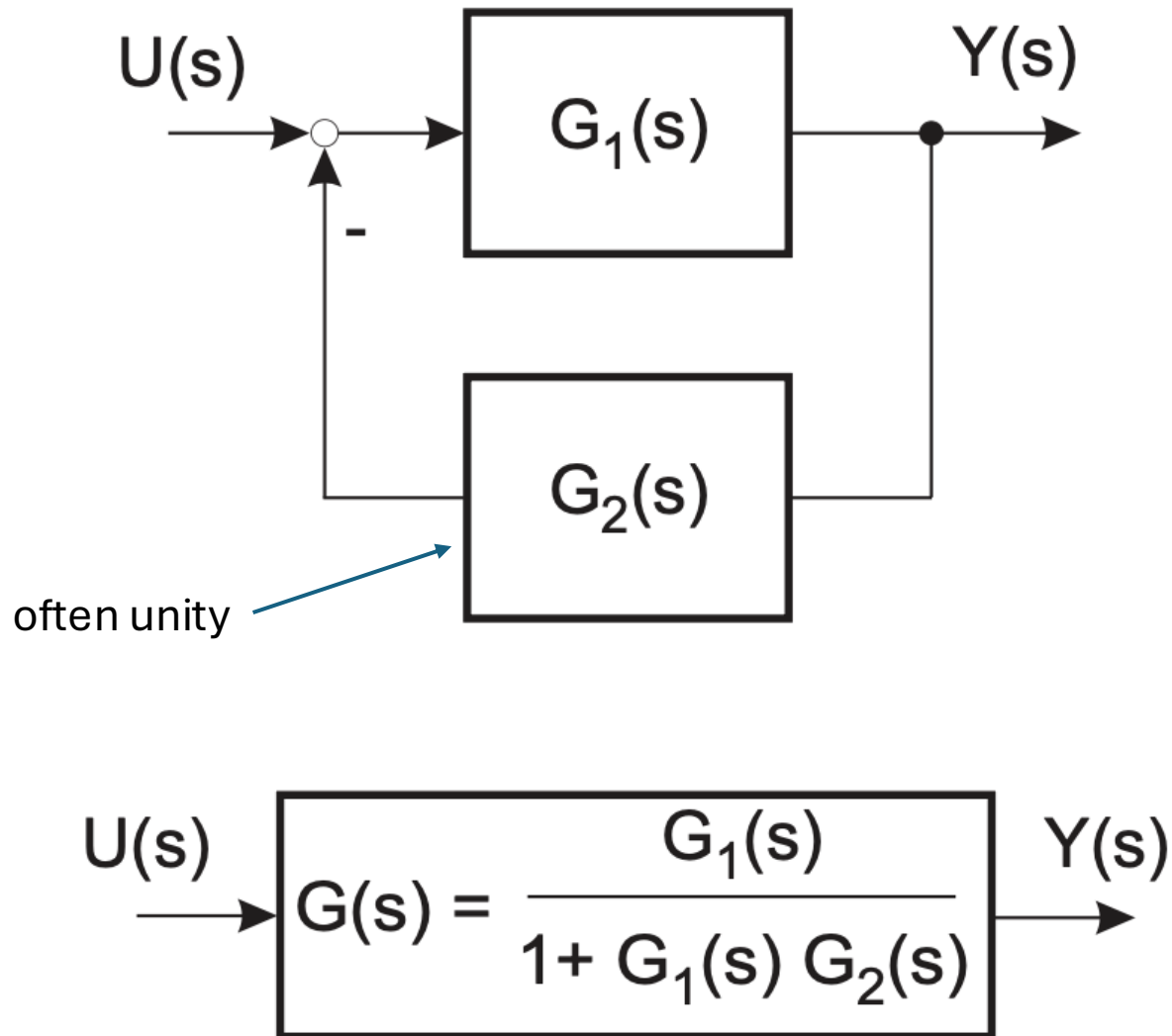
Example: PT2 PT1 PT1



Parallel Connection



Feedback Connection



$$Y(s) = G_1(s) \cdot E(s)$$
$$E(s) = U(s) - G_2(s) \cdot Y(s)$$

$$\Rightarrow Y(s) = G_1(s) \cdot (U(s) - G_2(s) \cdot Y(s))$$

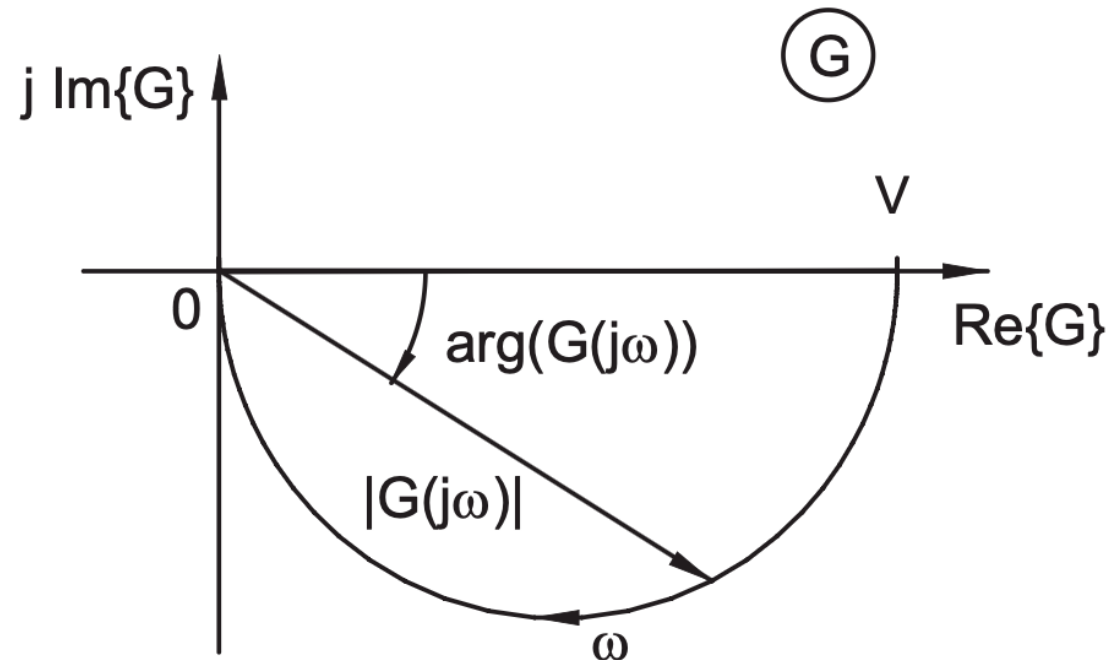
$$\Rightarrow Y(s) \cdot (1 + G_1(s) \cdot G_2(s)) = G_1(s) \cdot U(s)$$

$$\Rightarrow \frac{Y(s)}{U(s)} = \frac{G_1(s)}{(1 + G_1(s) \cdot G_2(s))}$$

Frequency Response

- Response to sinusoidal inputs: $G(j\omega)$
- Amplitude and phase as function of frequency ω

Nyquist diagram:

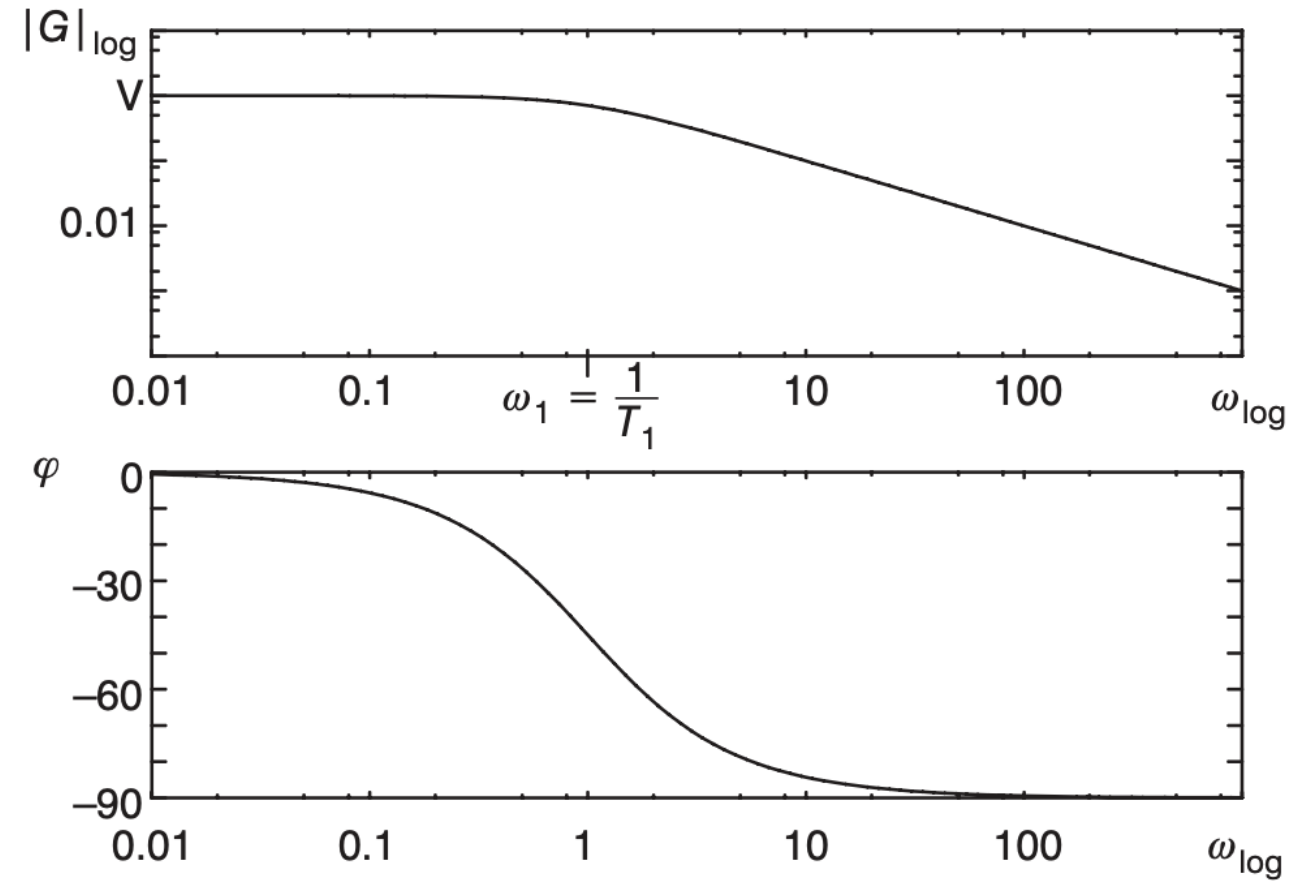


Bode Plots

1. Magnitude (dB) vs frequency
2. Phase vs frequency

Logarithmic frequency axis

Example: PT1



$$\text{dB} = 10 \cdot \log_{10} \left(\frac{P_2}{P_1} \right)$$

For amplitudes ($P \sim A^2$):

$$\text{dB} = 20 \cdot \log_{10}(|G(j\omega)|)$$

Bandwidth

Frequency range with good tracking (i.e., closed-loop system follows reference signal closely in shape, speed, and steady-state value)

Strong damping for higher frequencies (kind of filter on reference)

Typically consider frequency for which $|G(j\omega)|$ changes by -3 dB:
 $|G(j\omega_B)| = \frac{1}{\sqrt{2}} \cdot |G(0)|$ (amplitude drops to 70.7% of its DC value)

Indicates speed of system: higher bandwidth $\rightarrow$ faster response

But also less robust and higher noise amplification

Individual Elements of Control Loops

Polynomial form of differential equation / transfer function:

$$(a_n s^n + \dots + a_1 s + a_0) \cdot Y(s) = (b_m s^m + \dots + b_1 s + b_0) \cdot U(s)$$

→ Transfer functions of individual elements by setting a portion of the coefficients to zero

Proportional Element (P)

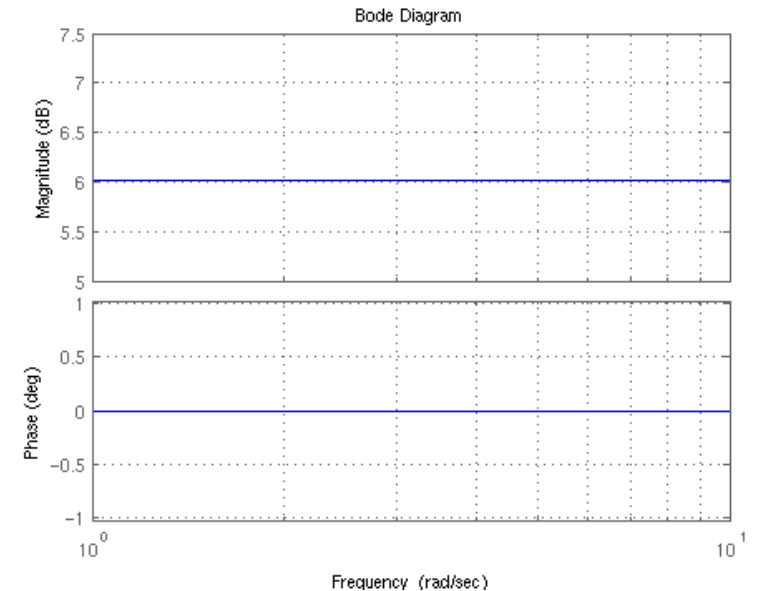
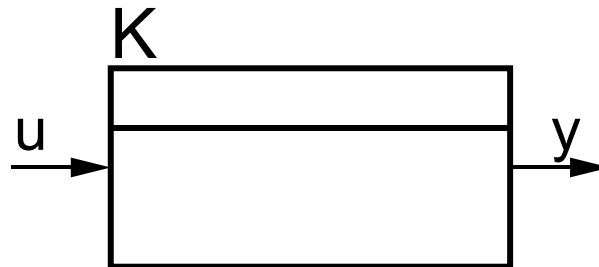
Consider $n = 0$:

$$y(t) = \frac{b_0}{a_0} \cdot u(t) \quad \text{or} \quad Y(s) = \frac{b_0}{a_0} \cdot U(s)$$

$$\frac{b_0}{a_0} = K_P$$

(just constant gain)

Step response:



Integral Element (I)

Consider $n = 1$, with $a_0 = b_1 = 0$:

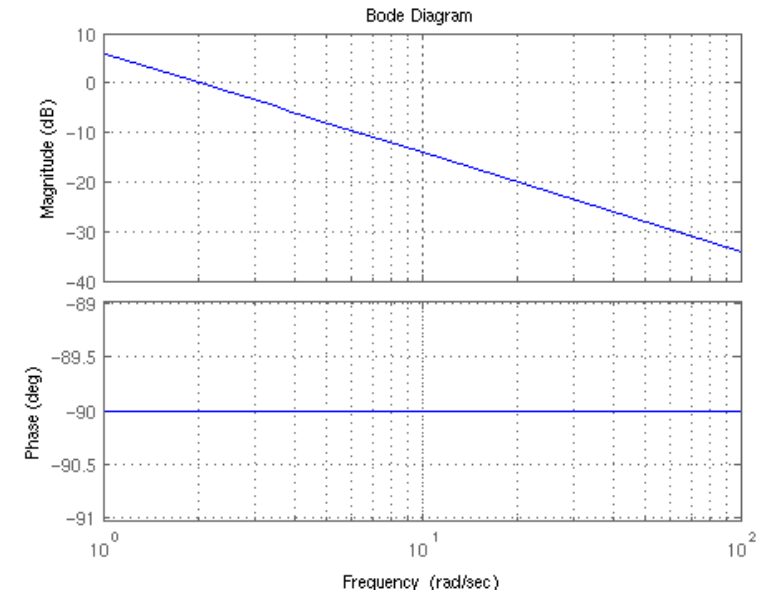
Pole at $s = 0$

$$\frac{dy(t)}{dt} = \frac{b_0}{a_1} \cdot u(t) \quad \text{or} \quad Y(s) = \frac{b_0}{a_1} \cdot \frac{1}{s} \cdot U(s)$$

$$y(t) = \frac{b_0}{a_1} \cdot \int u(t) dt$$

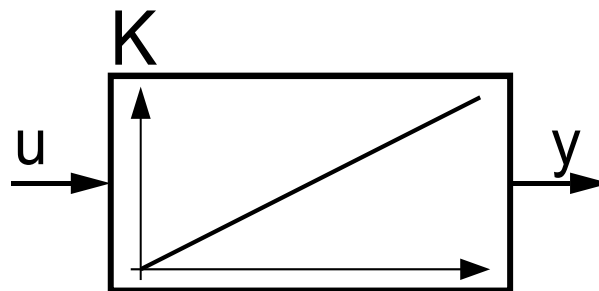
$$\frac{b_0}{a_1} = K_I = 1/T_I$$

(integrator gain or integral time constant)



Step response:

$$\int \text{const} dt = t$$



Energy storage,
memory

First-Order Lag Element (PT1)

Consider $n = 1$, with $b_1 = 0$:

$$\frac{a_1}{a_0} \cdot \frac{dy(t)}{dt} + y(t) = \frac{b_0}{a_0} \cdot u(t) \quad \text{or} \quad Y(s) \cdot \left(s \cdot \frac{a_1}{a_0} + 1 \right) = \frac{b_0}{a_0} \cdot U(s)$$

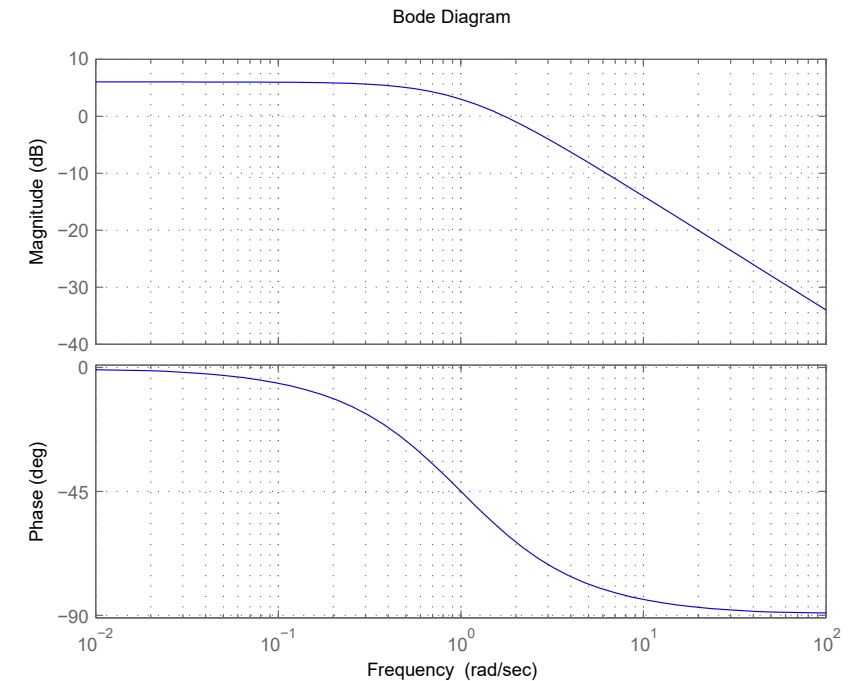
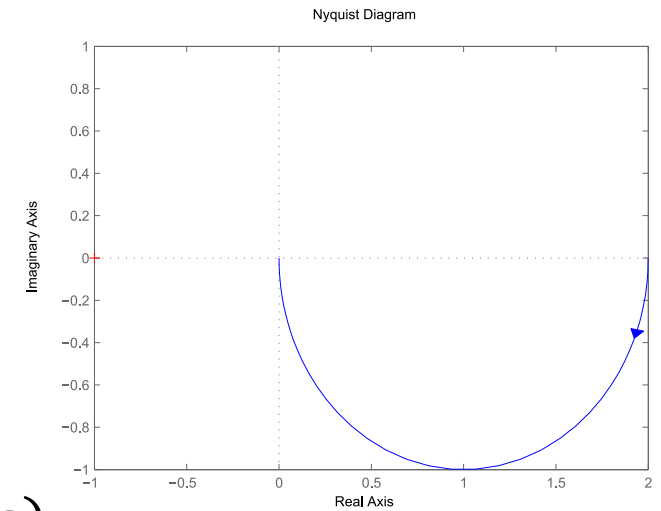
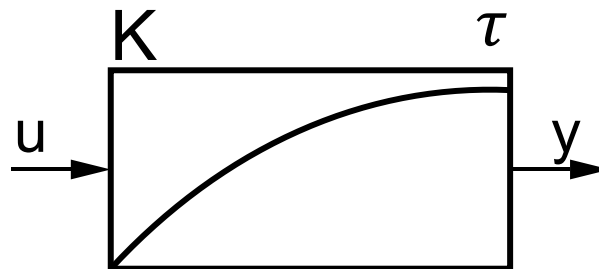
τ

K

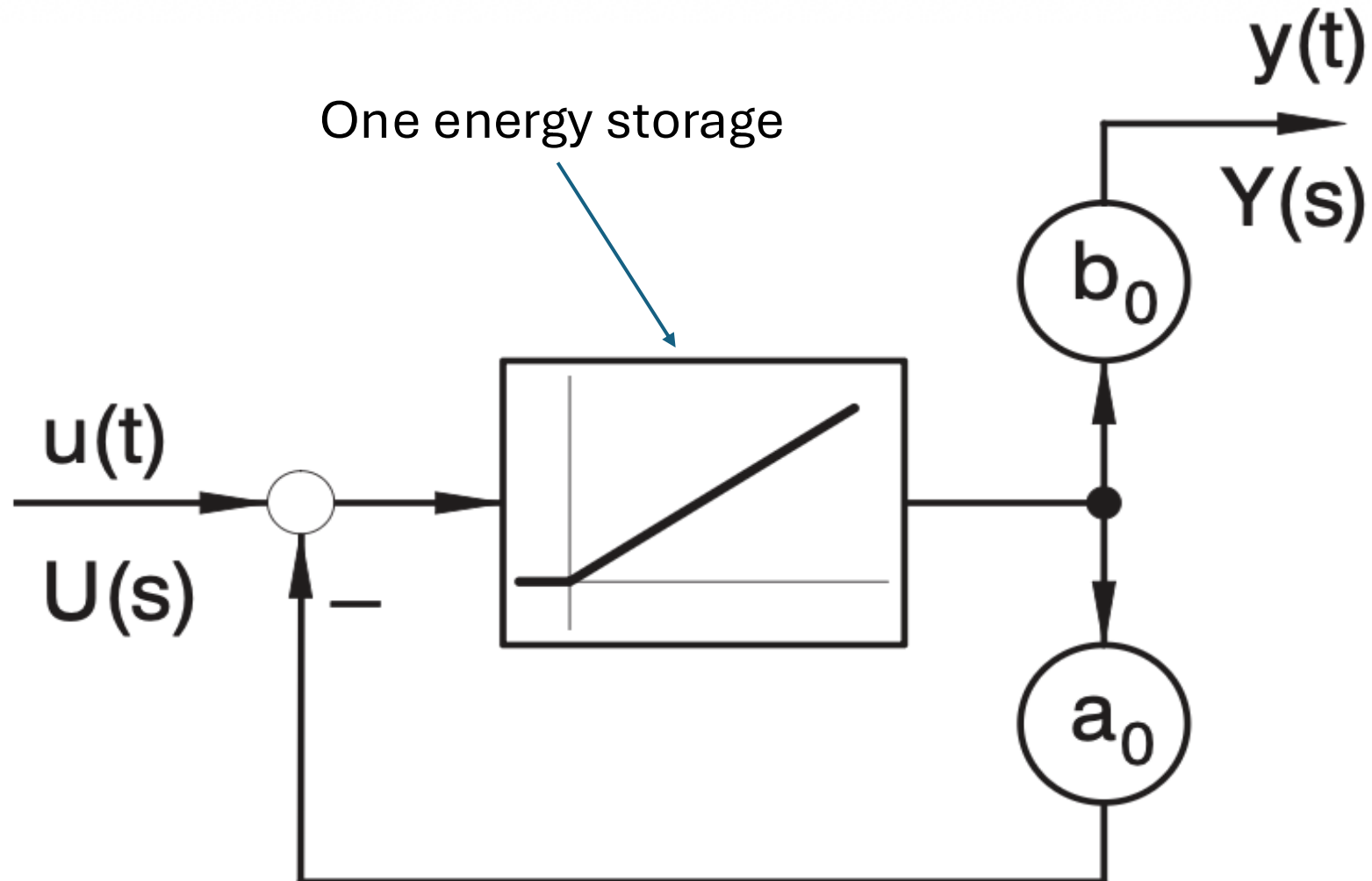
$$G(s) = \frac{Y(s)}{U(s)} = \frac{K}{1 + \tau s}$$

Pole at $s = -\frac{1}{\tau}$

Step response:



PT1 Signal Flow Diagram



Derivative Element (D)

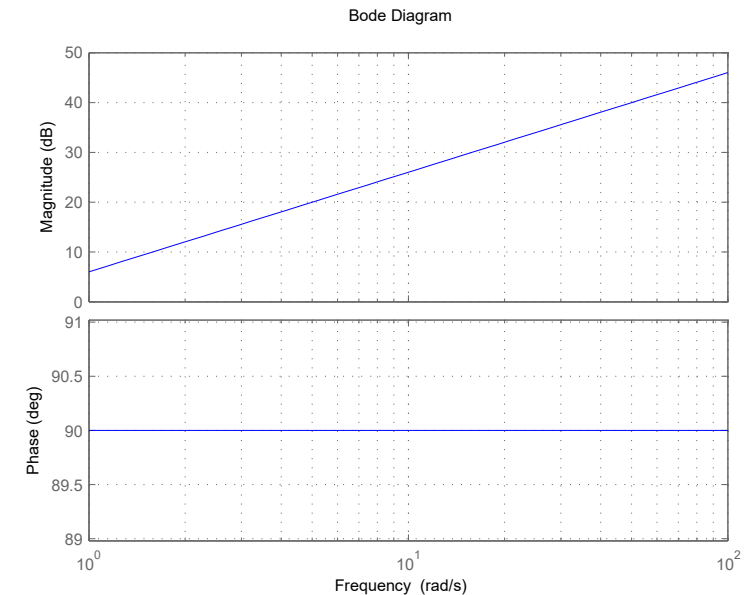
Consider $n = 1$, with $a_1 = b_0 = 0$:

$$y(t) = \frac{b_1}{a_0} \cdot \frac{du(t)}{dt} \quad \text{or} \quad Y(s) = \frac{b_1}{a_0} \cdot s \cdot U(s)$$

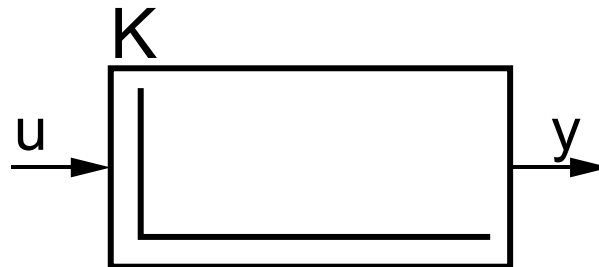
$$\frac{b_1}{a_0} = K_D = T_D$$

(differentiator/derivative gain or derivative time constant)

Amplifies high frequencies, i.e., noise
→ use filtered differentiator (DT1)



Step response:



physically unrealizable → damping

Filtered Differentiator (DT1)

Consider $n = 1$, with $b_0 = 0$:

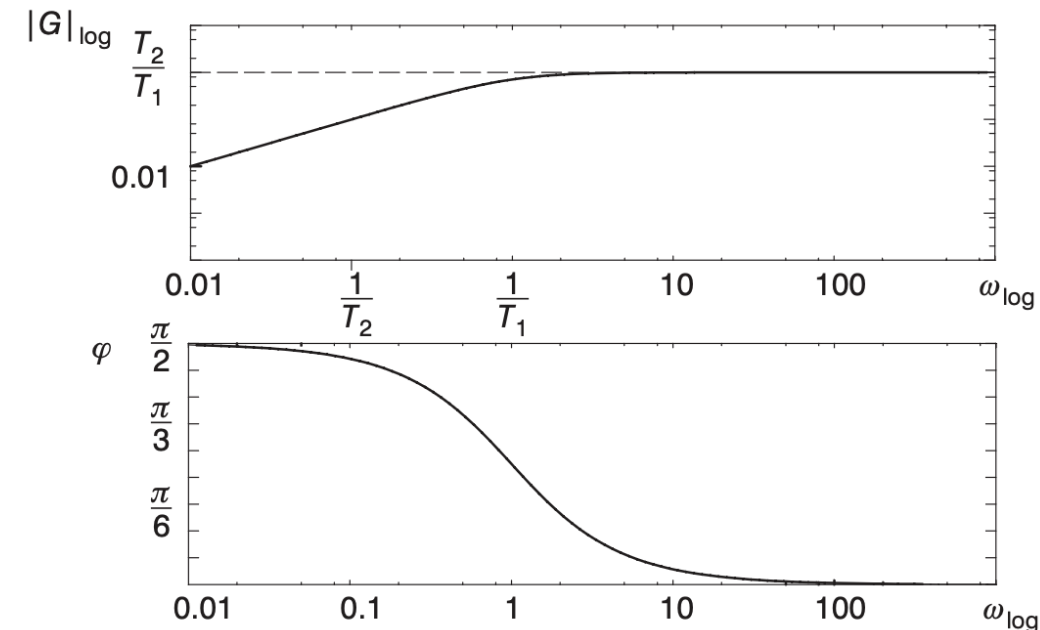
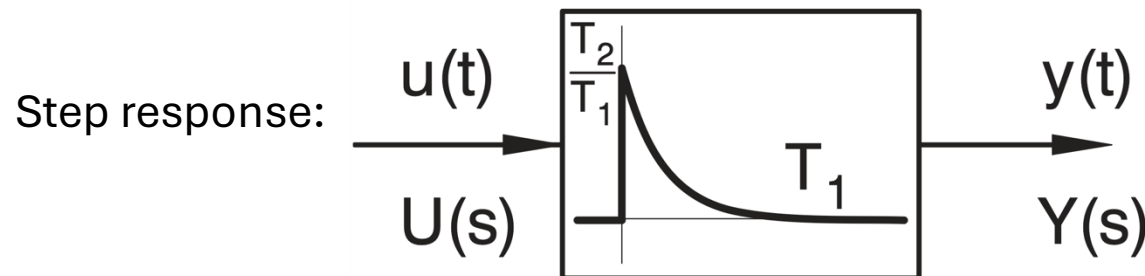
$$\frac{a_1}{a_0} \cdot \frac{dy(t)}{dt} + y(t) = \frac{b_1}{a_0} \cdot \frac{du(t)}{dt} \quad \text{or} \quad Y(s) \cdot \left(s \cdot \frac{a_1}{a_0} + 1 \right) = \frac{b_1}{a_0} \cdot s \cdot U(s)$$

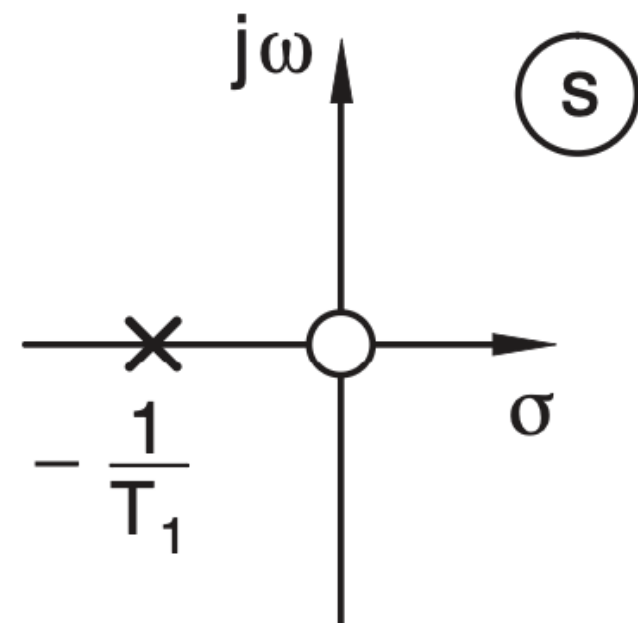
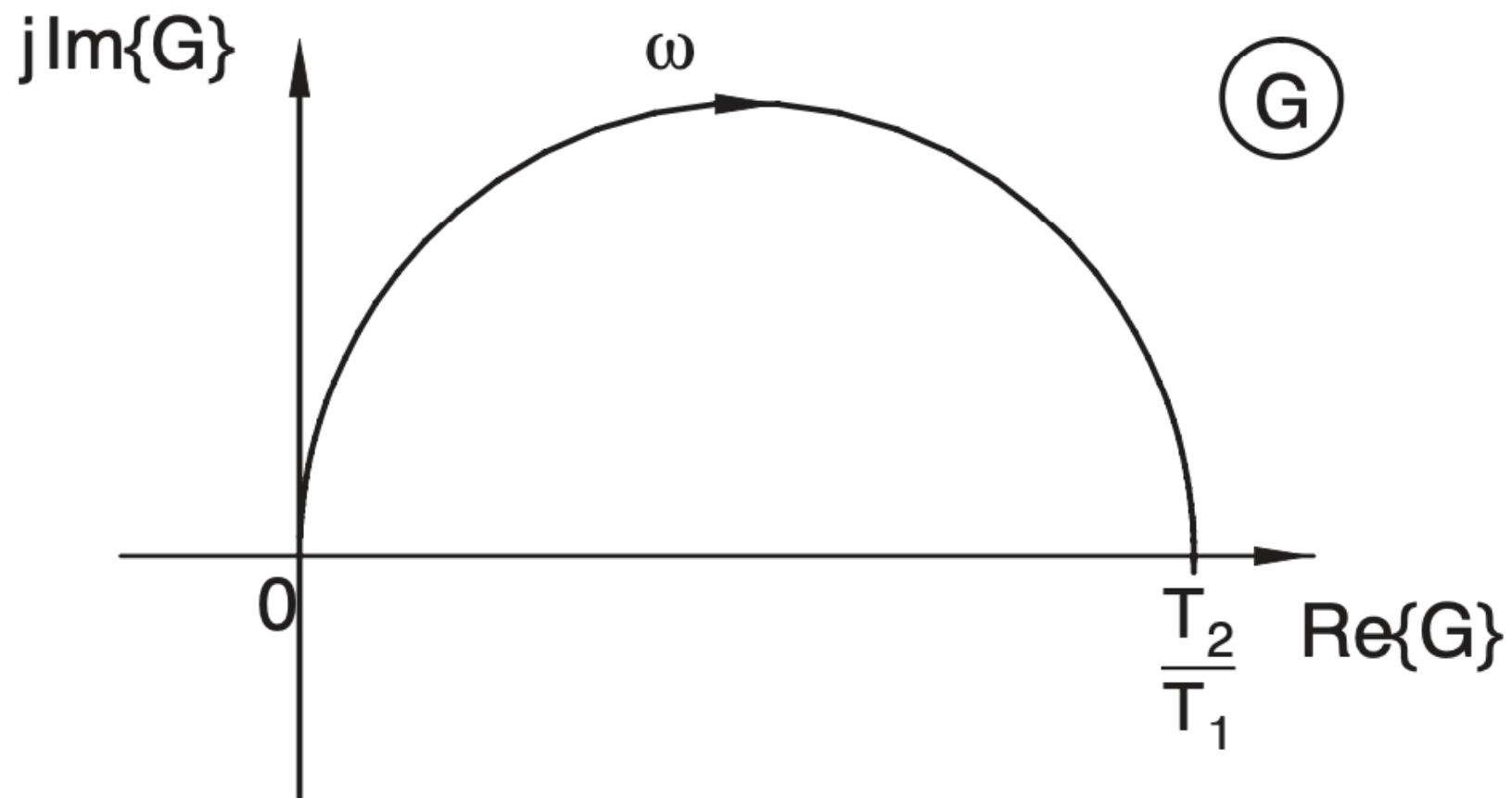
$\swarrow T_1$
 $\swarrow T_2$

$$G(s) = \frac{T_2 \cdot s}{T_1 \cdot s + 1}$$

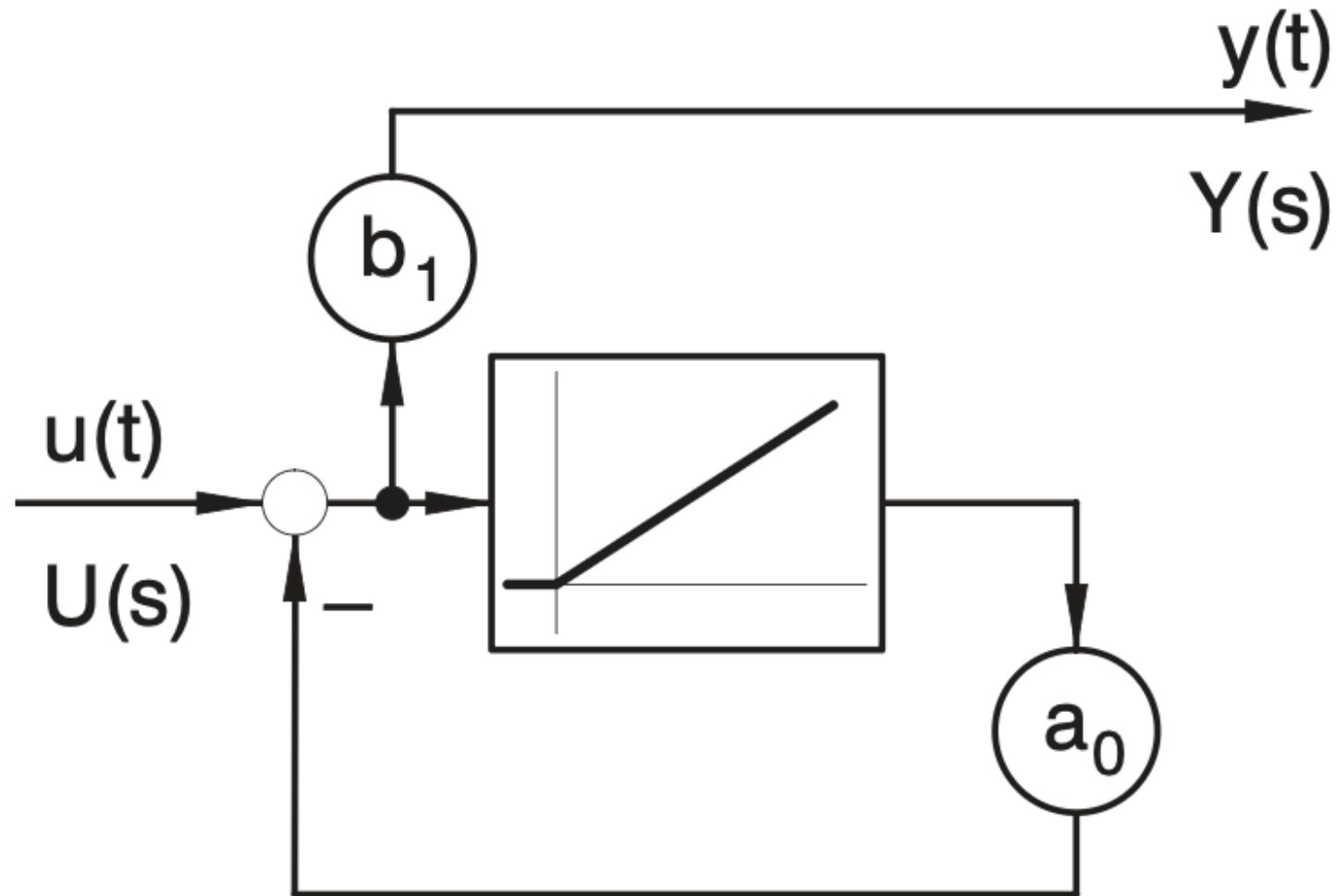
$$\lim_{T_1 \rightarrow 0} G(s) = T_2 \cdot s$$

→ ideal D element





DT1 Signal Flow Diagram



Second-Order Lag Element (PT2)

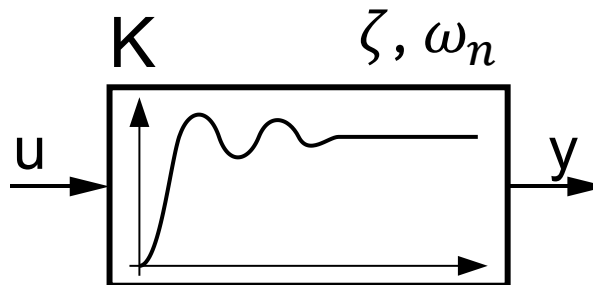
Consider $n = 2$, with $b_1 = b_2 = 0$:

$$\frac{1}{\omega_0^2} \frac{a_2}{a_0} \frac{d^2 y(t)}{dt^2} + \frac{2\zeta}{\omega_n} \frac{a_1}{a_0} \frac{dy(t)}{dt} + y(t) = \frac{b_0}{a_0} u(t) \quad K$$

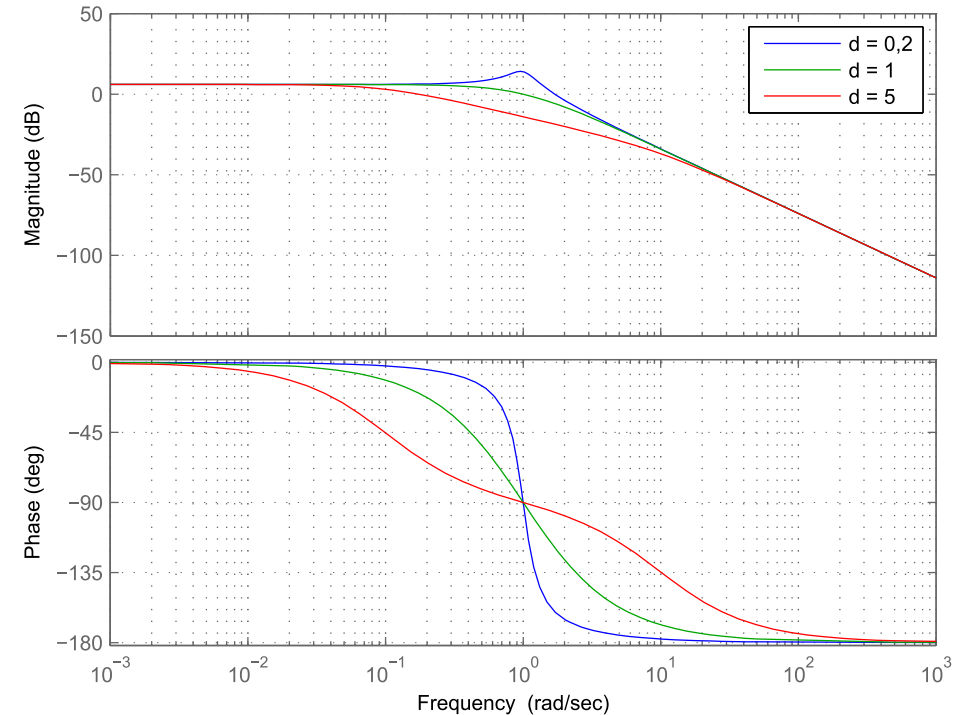
$$Y(s) \cdot \left(\frac{a_2}{a_0} s^2 + \frac{a_1}{a_0} s + 1 \right) = \frac{b_0}{a_0} \cdot U(s)$$

$$G(s) = \frac{K}{\frac{1}{\omega_n^2} s^2 + \frac{2\zeta}{\omega_n} s + 1}$$

Step response:



Bode Diagram



PT2 as Product of Two PT1 Elements

$G(s) = \frac{K}{\frac{1}{\omega_n^2}s^2 + \frac{2\zeta}{\omega_n}s + 1}$ can be written as product of two PT1 elements:

$$G(s) = \frac{K}{(\tau_1 s + 1)(\tau_2 s + 1)}$$

individual time constants

$$\Rightarrow \frac{1}{\omega_n^2}s^2 + \frac{2\zeta}{\omega_n}s + 1 = \tau_1\tau_2s^2 + (\tau_1 + \tau_2)s + 1$$

$$\Rightarrow \frac{1}{\omega_n^2} = \tau_1\tau_2, \quad \frac{2\zeta}{\omega_n} = (\tau_1 + \tau_2)$$

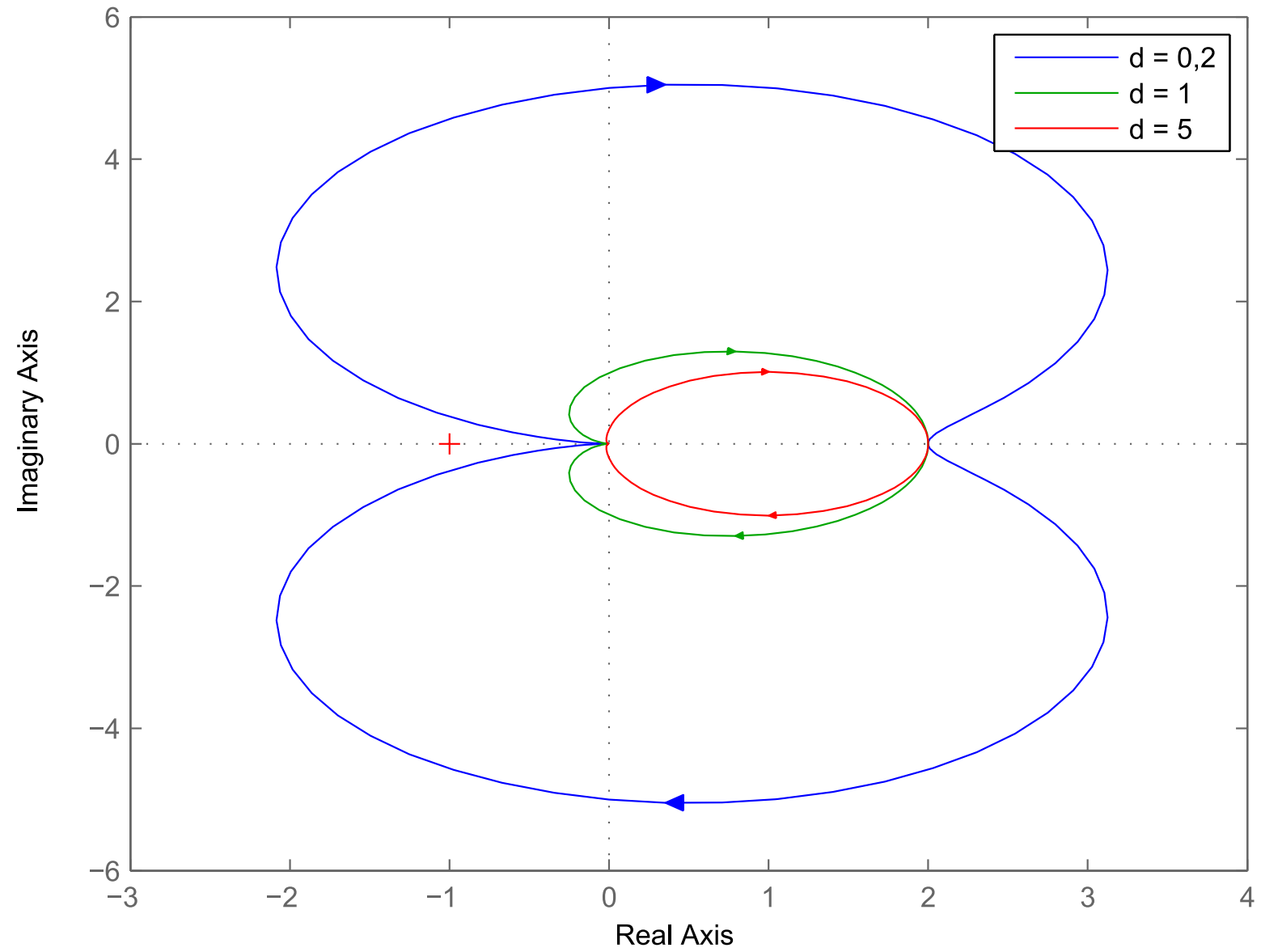
$$\Rightarrow \zeta = \frac{\tau_1 + \tau_2}{2\sqrt{\tau_1\tau_2}}$$

for real PT1 elements: $\tau_1 + \tau_2 \geq 2\sqrt{\tau_1\tau_2}$ (arithmetic-geometric mean inequality)

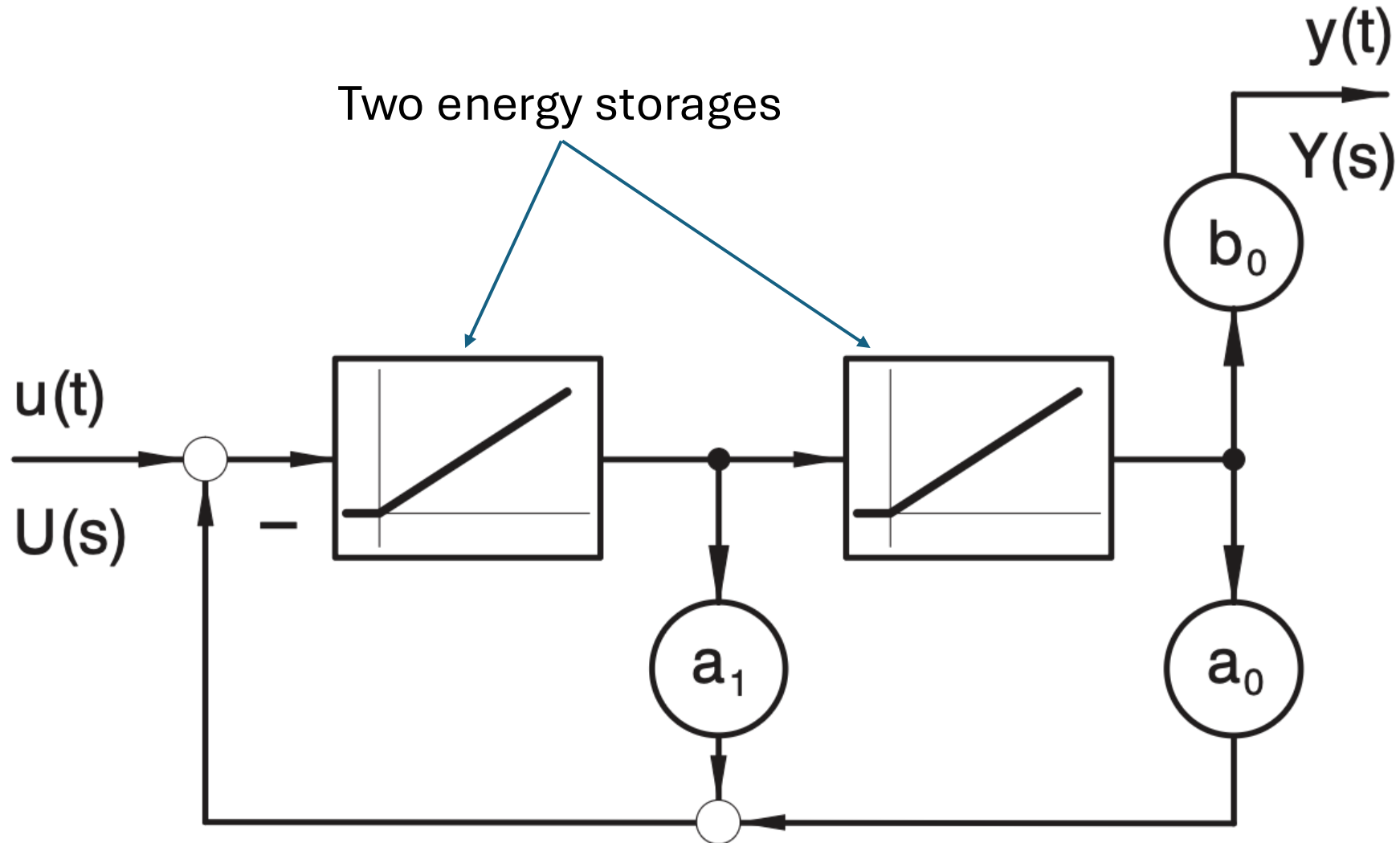
→ overdamped ($\zeta < 1$) or critically damped ($\zeta = 1$) for $\tau_1 = \tau_2$

fastest non-oscillatory response

Nyquist Diagram



PT2 Signal Flow Diagram



All-Pass Systems

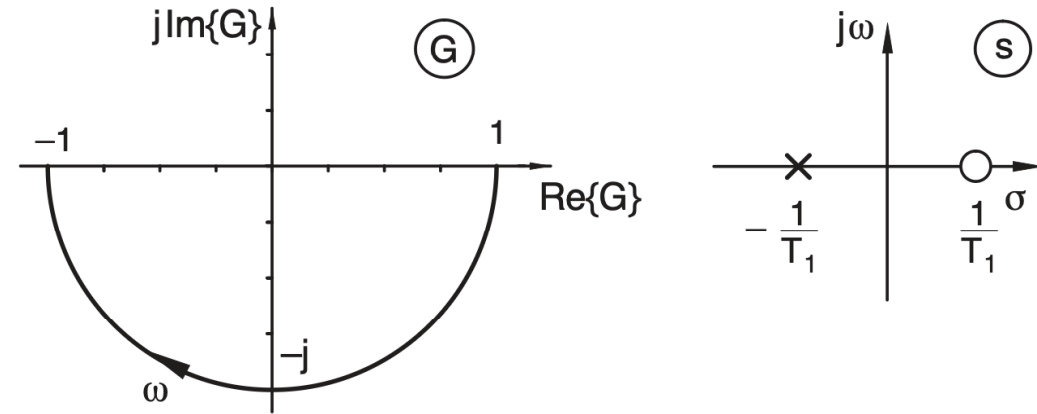
Magnitude constant for all frequencies
 $(|G(j\omega)| = 1)$, but phase depends on frequency

coefficients: $b_k = (-1)^{n-k} a_{n-k}$

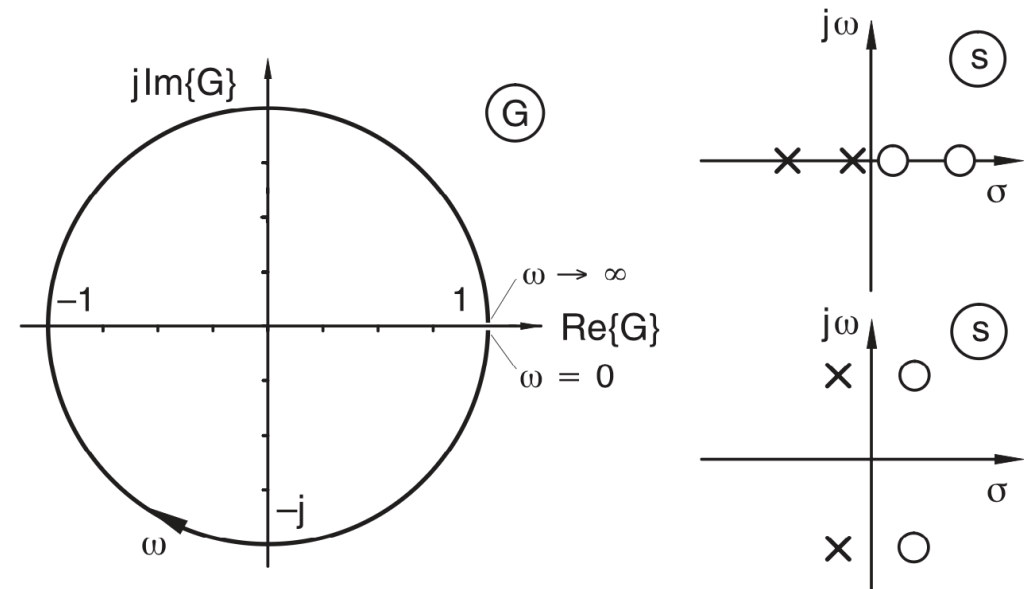
→ Zeros are mirror images of the poles with respect to the imaginary axis.

Important in signal processing, e.g., for runtime alignment

first-order all-pass: $G(s) = \frac{s-a}{s+a}$



second-order all-pass: $G(s) = \frac{a_2 s^2 - a_1 s + a_0}{a_2 s^2 + a_1 s + a_0}$



Systems with Dead Time

The transfer function of a system can be extended by multiplying it with the exponential term that represents a time delay:

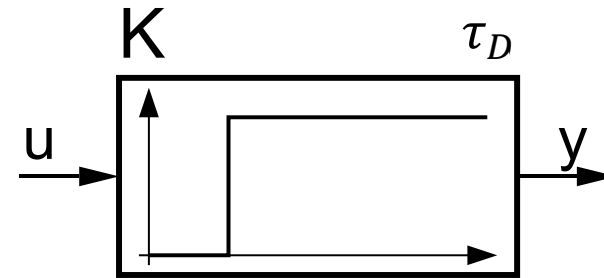
$$G(s) = G_0(s) \cdot e^{-s \cdot \tau_D}$$

original transfer function time delay (aka dead time)

Example: PT1 with dead time

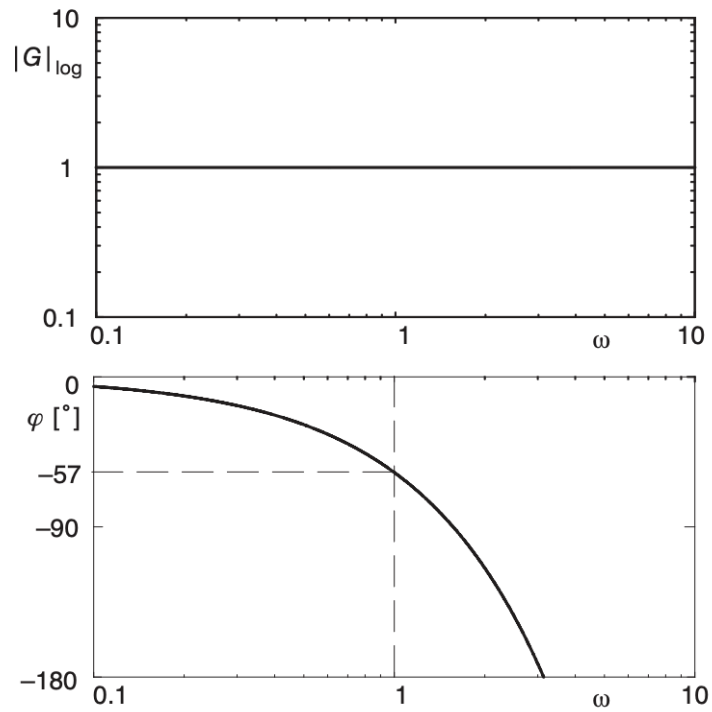
$$G(s) = \frac{K}{1 + \tau \cdot s} \cdot e^{-s \cdot \tau_D}$$

Step response (P element with dead time):

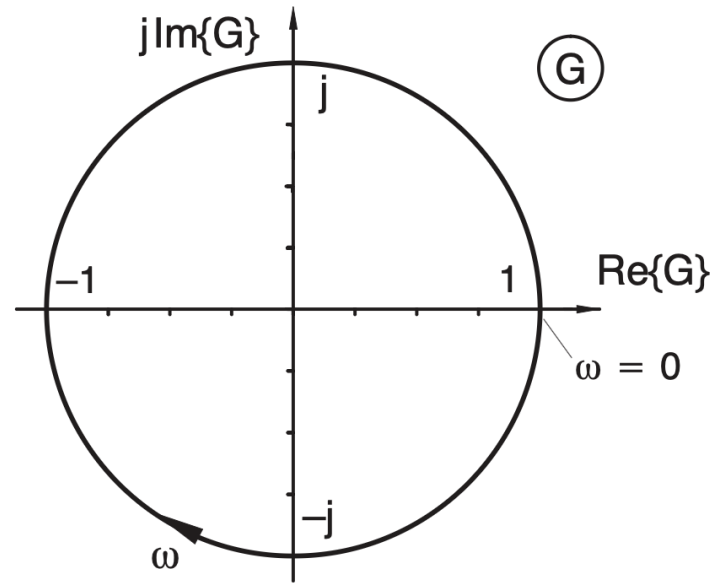


Mainly suitable for frequency response analysis (e.g., Bode plots)

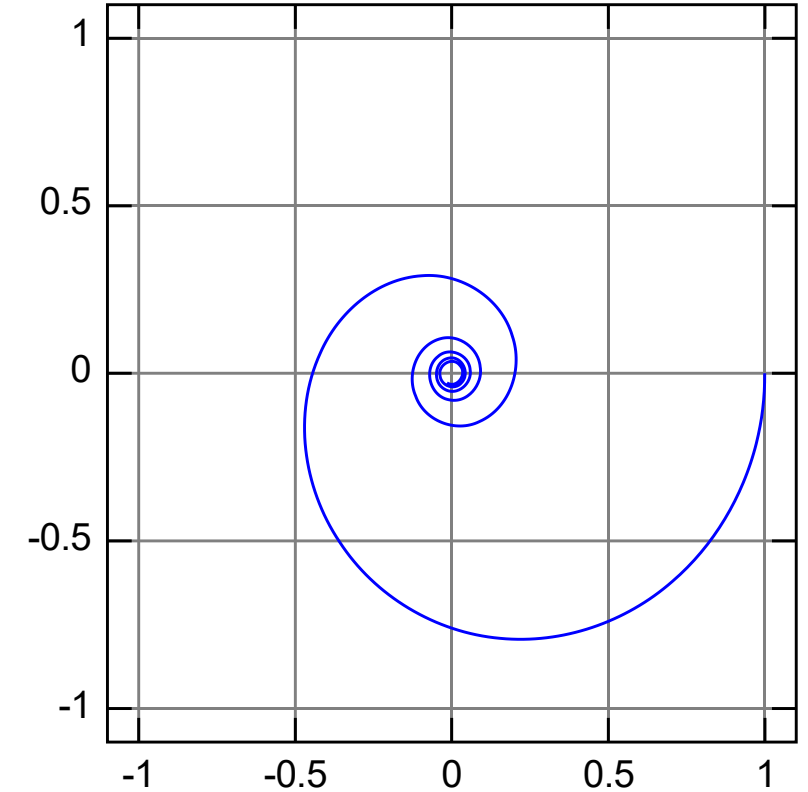
Bode diagram for system with dead time:



Nyquist diagram for system with dead time:



Nyquist diagram for PT1 with dead time:



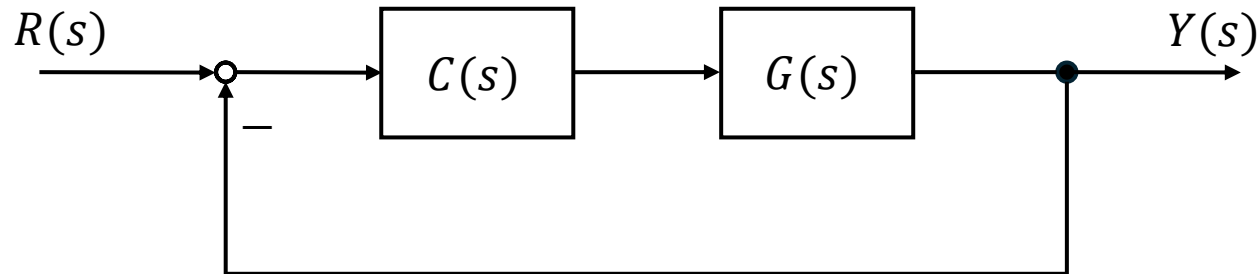
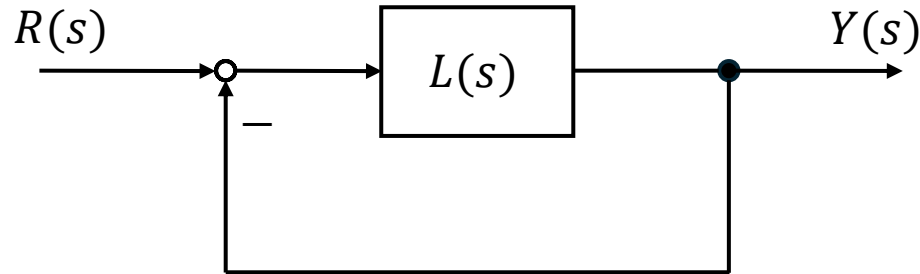
Control Loop

$$Y(s) = L(s) \cdot (R(s) - Y(s))$$

$$\Rightarrow Y(s) \cdot (1 + L(s)) = L(s) \cdot R(s)$$

$$\Rightarrow \frac{Y(s)}{R(s)} = \frac{L(s)}{1 + L(s)}$$

closed-loop transfer function
for negative feedback



$$\frac{Y(s)}{R(s)} = \frac{C(s) \cdot G(s)}{1 + C(s) \cdot G(s)}$$

$G(s)$ above combines controller $C(s)$ and plant $G(s)$

PID Controller Transfer Function

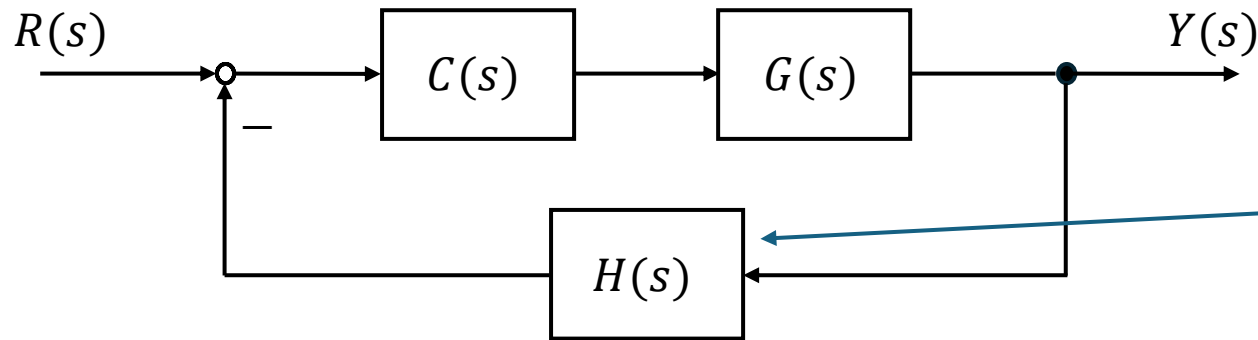
$$C(s) = K_p + \frac{K_i}{s} + K_d \cdot s$$

- K_p : increases loop gain $\rightarrow$ faster response, smaller steady-state error, may increase overshoot
- K_i : adds integrator $\rightarrow$ eliminates steady-state error, may reduce stability margin
- K_d : adds damping $\rightarrow$ reduces overshoot, improves transient behavior

Controller design:

- Shape the closed-loop poles to obtain the desired dynamics
- Achieved by selecting and tuning controller parameters (move the closed-loop poles)

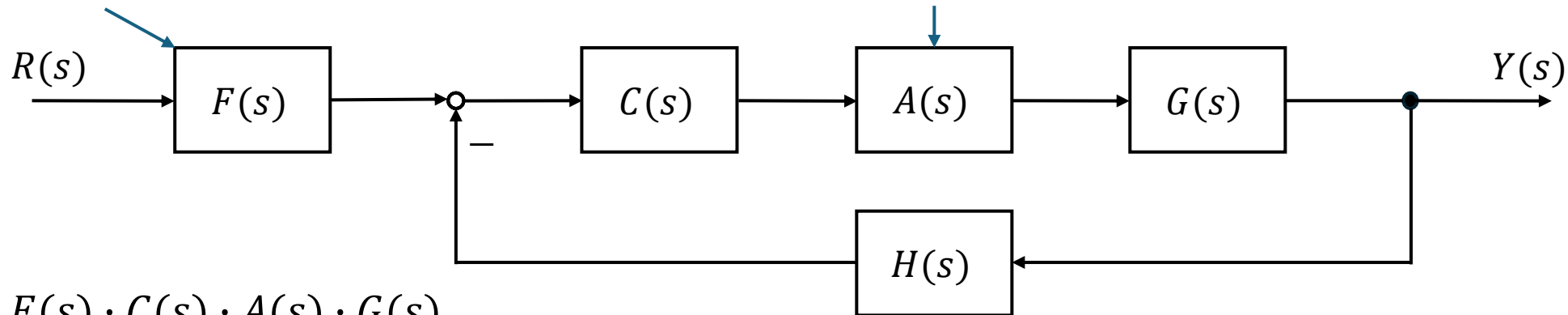
Hidden Assumptions



$$\frac{Y(s)}{R(s)} = \frac{C(s) \cdot G(s)}{1 + C(s) \cdot G(s) \cdot H(s)}$$

sensor dynamics: If sensor is fast, one can assume unity feedback.

reference prefilter: If reference shaping doesn't affect stability, one can treat it separately.

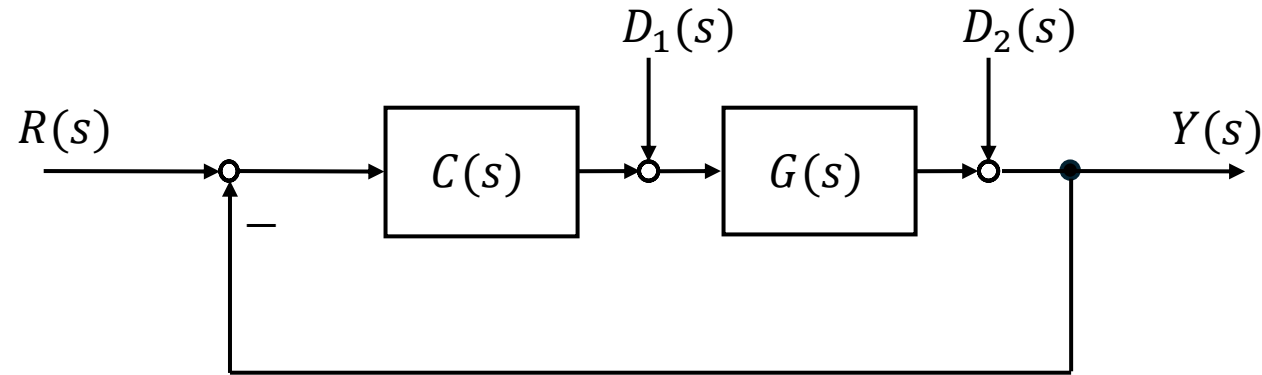


actuator dynamics: If actuator bandwidth $\gg$ closed-loop bandwidth, one can ignore it.

$$\frac{Y(s)}{R(s)} = \frac{F(s) \cdot C(s) \cdot A(s) \cdot G(s)}{1 + C(s) \cdot A(s) \cdot G(s) \cdot H(s)}$$

Always model what can affect poles

Disturbances



$$Y(s) = \frac{C(s) \cdot G(s)}{1 + C(s) \cdot G(s)} \cdot R(s) + \frac{G(s)}{1 + C(s) \cdot G(s)} \cdot D_1(s) + \frac{1}{1 + C(s) \cdot G(s)} \cdot D_2(s)$$

Output as superposition of input and disturbances